

- Q1. Which quantum numbers depend on the angular wavefunction? [1]
- Q2. Give the quantum numbers defined by the angular wavefunction for a dz^2 orbital. [2]
- Q3. Based on the building-up principle, or the Aufbau principle, write down the shells and subshells arranged in order of increasing energy for the occupation of electrons in multielectron atoms. [2]
- Q4. In a hydrogen-like atom, which of the 4s-electron and 3d-electron has lower energy and why? [2]
- Q5. What is the degree of degeneracy of a hydrogen-like atom's second excited state? [2]
- Q6. Comment on the angular dependence of the angular wavefunctions, $Y_{0,0}$ and $Y_{1,0}$. [4]
- Q7. What is the eigen value when the z-component of the angular momentum operator (L_z) operates on the Spherical Harmonics eigenfunction, $Y_l^m(\theta, \varphi)$. [2]
- Q8. Calculate the radial distance, $r_{\max}$, at which the density probability acquires its maximum value for the probability density for a given wavefunction given by [4]
- $$P(r) = 4 \left(\frac{Z}{a}\right)^3 \exp\left\{-\frac{2Zr}{a}\right\} r^2.$$
- Q9. Write the *hamiltonian* for the hydrogenic atom. [3]
- Q10. A p orbital is found to have one node in addition to the nodal plane that bisects the lobes. What would you predict to be the value of n ? [2]
- Q11. If an s orbital has two nodes, what is the value of n ? [2]
- Q12. Two angular wavefunctions are given below. **Predict** the atomic orbitals and **explain** how you made your decision. [3+3]
- (i) $\left(\frac{3}{4\pi}\right)^{1/2} \cos\theta$
- (ii) $\left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\cos\varphi$
- Q13. Is each set of quantum numbers allowed? Explain your answers. [6]
- (i) $n = 2; l = 1; m_l = 2; m_s = +1/2$
- (ii) $n = 3, l = 0; m_l = -1; m_s = -1/2$
- (iii) $n = 2; l = 2; m_l = 1; m_s = +1/2$
- Q14. What quantum number defines each of the following? [8]
- (i) the overall shape of an orbital
- (ii) the orientation of an electron with respect to a magnetic field
- (iii) the orientation of an orbital in space
- (iv) the average energy and distance of an electron from the nucleus

Q15. What value of l corresponds to a d subshell? How many orbitals are in this subshell? [2]

Q16. How many subshells and orbitals are found within the principal shell $n = 6$? How do these orbital energies compare with those for $n = 4$? [4]

Q17. The periodic table is divided into blocks. Identify each block and explain the principle behind the divisions. Which quantum number distinguishes the horizontal rows? [3,1]

Q18. Compare the first ionization energies of the consecutive members of the group 15 and group 16 elements. Explain. [4]

Q19. Most of the first row transition metals can form at least two stable cations, for example iron(II) and iron(III). In contrast, scandium and zinc each form only a single cation, the Sc^{3+} and Zn^{2+} ions, respectively. Use the electron configuration of these elements to provide an explanation. [2]

Q20. What four variables are required to fully describe the position of any object in space? In quantum mechanics, one of these variables is not explicitly considered. Which one? [2,1]

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